TECHNICAL REQUIREMENTS

BUSINESS MANAGEMENT TOOLS FOR MSMEs

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Field	Value
Document Version	1.1 Phase 1 (Pilot)
Date	March 2026
Client	Trident Aliska Digital Tech Ghana LTD
Programme	GRA MSME Tax & Business Management Programme
Status	Active in development

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1. Project Context

1.1 Background

Trident Aliska Digital Tech Ghana LTD has entered a Memorandum of Understanding (MOU) with the Ghana Revenue Authority (GRA) to deploy the Modified Tax and Business Management Tools for MSMEs Programme, a pilot initiative targeting informal sector traders at Makola Market and Accra CBD.

The programme provides market traders with:

- A point-of-sale device (Sunmi Android POS) to record transactions
- Automated Ghana tax calculation on every sale
- GRA-compliant invoice and receipt printing
- Basic business management tools (inventory, loans, savings, expenses)
- A pathway to formal tax registration and compliance

1.2 Scope of This Document

This TRD covers the technical requirements for three software components:

Component	**Repository**	**Responsible**
Backend REST API	triliska-backend	Vietnam/India team (shared, Trident Aliska admin)
Flutter POS App	triliska-pos	Vietnam/India team (shared, Trident Aliska admin)
GRA Middleware	triliska-middleware	Aliska (Trident Aliska Support)

1.3 Target Users

User Type	**Description**	**Primary Interface**
MSME Owner	Market trader using the Sunmi POS daily	Flutter app on Sunmi
Programme Staff	• NSP/YEA Technical Field Officers	Web browser + Sunmi

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User Type	Description	Primary Interface
	• Zonal Heads	
Project Sponsor & Affiliates	GRA / Ministry of Trade	Web browser (read only)
Project Implementor	Trident Aliska Digital Tech	Web browser (full access)

1.4 Target Hardware Phase 1

Device	Model	Screen	Printer	Notes
Sunmi V3 PLUS	Android POS	5.99"	58mm thermal	Primary pilot device
Sunmi V2 PRO	Android POS	5.99"	80mm thermal	Secondary; has barcode scanner

Both devices run Android with 2–4GB RAM. The Flutter app must run cleanly on these devices.

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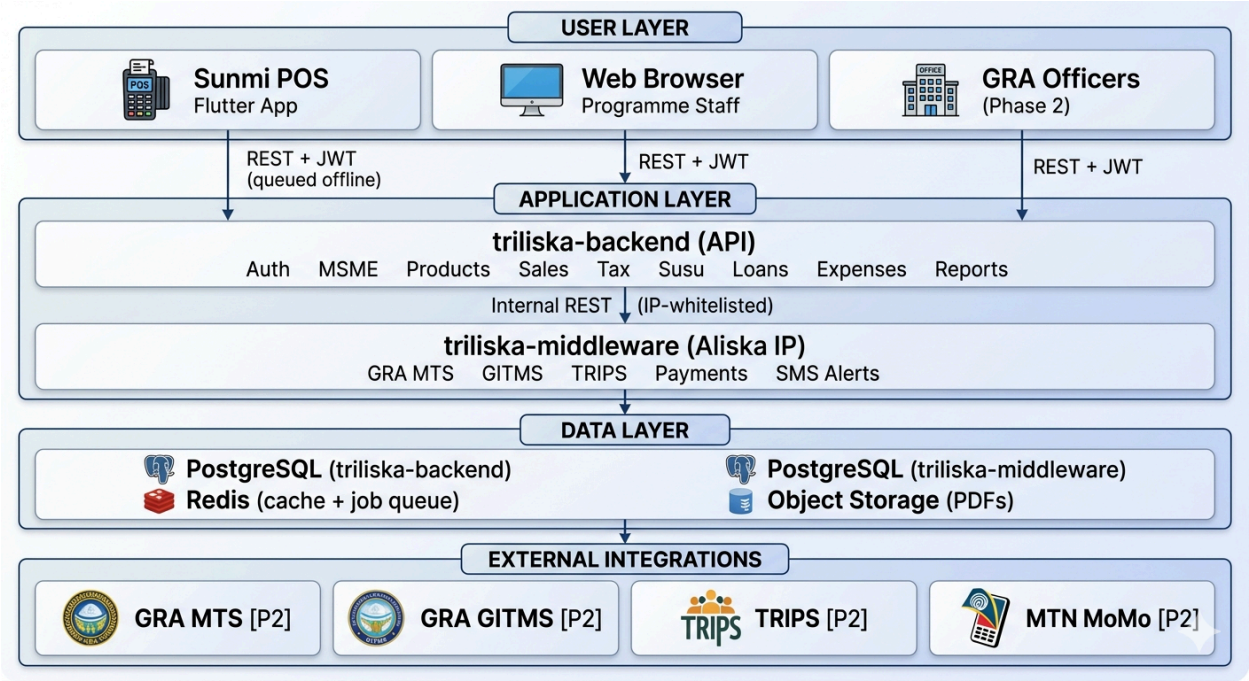
2. System Overview

2.1 Architecture Diagram



2.2 Technology Stack Decisions

Decision	Choice	Rationale
Backend language	Python (FastAPI) or TypeScript (Node/Fastify)	**Decide in Day 1 meeting based on team capability**
Database	PostgreSQL 15	ACID compliance, NUMERIC type for money, non-negotiable
Flutter	Flutter 3.x	Only viable option for Sunmi hardware SDK access
Middleware	Python / FastAPI	PDF generation, Celery scheduler, GRA integration

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Decision	Choice	Rationale
Auth	JWT (access + refresh tokens)	Stateless, works offline-first
Money	Decimal (Python) / decimal.js (Node) / Decimal (Dart)	Never floats
IDs	ULID	Sortable, no sequential leakage

2.3 Environments

Environment	Purpose	URL
Development	Local developer machines	localhost:8000
Staging	Integration testing before pilot	https://staging-api.triliska.com
Production	Live pilot	https://api.triliska.com

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3. Component 1 Backend API

3.1 Overview

The backend is a REST API that serves both the Flutter POS app and the Programme Staff web interface. It is the single source of truth for all business data.

Base URL: https://api.triliska.com/api/v1
Protocol: HTTPS only, no HTTP in any environment
Auth: JWT Bearer tokens
Content-Type: application/json

3.2 Module 1 Authentication [P1]

Requirements

ID	Requirement	Priority
AUTH-01	System shall authenticate users with username and password	P1
AUTH-02	Successful login returns a JWT access token (24hr expiry) and refresh token (30 days)	P1
AUTH-03	Refresh token endpoint issues a new access token without re-login	P1
AUTH-04	Logout endpoint invalidates the refresh token	P1
AUTH-05	All endpoints except /auth/login and /health require a valid JWT	P1
AUTH-06	Expired tokens return HTTP 401 with error code TOKEN_EXPIRED	P1
AUTH-07	Passwords stored as bcrypt hashes never plain text	P1
AUTH-08	Login request includes device_id for POS device tracking	P1
AUTH-09	Failed login after 5 attempts locks account for 15 minutes	P2

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ID	Requirement	Priority
AUTH-10	Multi-factor authentication for admin and district_officer roles	P2

Roles and Permissions

Role	Data Access	Can Create	Can Void
msme_owner	Own MSME only	Sales, expenses	Nothing
programme_staff	All MSMEs in their zone/district	MSMEs, sales	Sales (own zone)
district_officer	All MSMEs in district	Nothing	Nothing
admin	Everything	Everything	Everything

3.3 Module 2 MSME Registration & Profile [P1]

Requirements

ID	Requirement	Priority
MSME-01	Programme staff can register a new MSME with business name/registration, owner name, Ghana Card, phone, TIN, sector, and address	P1
MSME-02	System assigns a unique programme reference number on registration (format: TRL-{DISTRICT}-{SEQ})	P1
MSME-03	MSME profile stores Income Tax status:	P1

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ID	Requirement	Priority
	not_registered, flat_rate, or standard	
MSME-04	MSME profile stores GPS coordinates (lat/lng) for market location	P1
MSME-05	Programme staff can list all MSMEs in their zones, by industry and region, with filters	P1
MSME-06	MSME profile can be updated via PATCH (partial update)	P1
MSME-07	When MSME cumulative annual turnover crosses GHS 500,000 or GHS 750,000 (3%), system flags the threshold does NOT auto-change Income Tax status	P1
MSME-08	Income Tax status changes require programme_staff or admin role and are logged to audit trail	P1
MSME-09	MSME profile includes aggregate stats: total sales this month, transaction count, last transaction date	P1
MSME-10	MSME registration database supports national-scale coverage (2,000,000+ MSMEs). Year 1 target: 540,000	P2

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ID	Requirement	Priority
MSME-11	Bulk MSME import from CSV for zone/district-level onboarding	P2

Data Fields MSME Profile

Field	Type	Required	Notes
id	ULID string	Auto	msme_{ulid}
business_name	string	Yes	
owner_name	string	Yes	
owner_phone	string	Yes	E.164 format
owner_national_id	string	Yes	Ghana Card number
tin	string	No	GRA Tax ID Number
business_type	enum	Yes	sole_proprietor, partnership, company
sector	enum	Yes	See sector list below
address	string	Yes	
gps_lat	decimal	Yes	
gps_lng	decimal	Yes	
district	string	Yes	
zone	string	No	Sub-district zone
vat_status	enum	Yes	not_registered, flat_rate, standard

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Field	Type	Required	Notes
annual_turnover_band	enum	Yes	below_200k, 200k_to_500k, above_500k
preferred_language	enum	Yes	en, tw, ga, ee, ha, dag
programme_ref	string	Auto	TRL-ACC-0042 format
registration_date	datetime	Auto	
status	enum	Auto	active, inactive, suspended

Sector values: retail_food, retail_general, services, manufacturing, agriculture, transport, hospitality, other

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3.4 Module 3 Products & Inventory [P1]

Requirements

ID	Requirement	Priority
INV-01	MSME can create products with name, SKU, barcode, category, unit, selling price, cost price, and tax applicability	P1
INV-02	Each product has a current stock quantity and a low-stock threshold	P1
INV-03	Stock is automatically reduced when a sale is recorded	P1
INV-04	Stock movements (restock, wastage,	P1

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ID	Requirement	Priority
	adjustment) can be manually recorded with movement type, quantity, and cost	
INV-05	Inventory report shows all products with current stock, stock value, and low-stock alerts	P1
INV-06	System prevents a sale if stock would go below zero returns HTTP 422	P1
INV-07	Product list is returned in /sync/master-data for offline caching on Flutter app	P1
INV-08	Barcode field on product for scanner integration (V2 PRO)	P1
INV-09	Product categories and units are configurable (not hardcoded)	P1
INV-10	Import products from CSV (bulk product setup)	P1
INV-11	Import opening stock from CSV	P1
INV-12	Inventory forecasting (reorder suggestions based on sales velocity)	P2
INV-13	Supplier management (record purchase orders and supplier details)	P2
INV-14	Product variations (size, colour, weight)	P2

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3.5 Module 4 Sales [P1]

This is the most critical module. Every trader interaction flows through this.

Requirements

ID	Requirement	Priority
SALE-01	Record a sale with one or more line items, payment method, and optional customer	P1
SALE-02	Sale automatically calculates Ghana tax based on MSME's vat_status	P1
SALE-03	Sale response includes full tax breakdown (VAT, NHIL, GETFund, COVID levy)	P1
SALE-04	Sale generates a unique GRA-compliant invoice number (format: INV-TRL-{DISTRICT}-{MSME_REF}-{SEQ})	P1
SALE-05	Sale accepts a client_ref (device-generated UUID) for idempotency duplicate client_ref returns HTTP 200 with original sale, never creates a duplicate	P1
SALE-06	recorded_at field stores device timestamp; created_at stores server receipt time	P1
SALE-07	Sale can be voided by programme_staff or	P1

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ID	Requirement	Priority
	admin stock is restored on void	
SALE-08	Credit sales supported record amount owed, track payments against the sale	P1
SALE-09	Sale creation and all stock deductions happen in a single atomic database transaction	P1
SALE-10	Customer can be optionally linked to a sale from the customer list	P1
SALE-11	Sales list endpoint supports filtering by date range, sale type, and customer	P1
SALE-12	Receipt endpoint returns formatted lines array for Sunmi thermal printer (58mm and 80mm)	P1
SALE-13	Payment method supports: cash, cards, credit, MTN MoMo, Vodafone Cash, AirtelTigo, bank transfer	P1
SALE-14	For cash payments, response includes change_due calculated from amount_tendered	P1
SALE-15	Discount can be applied at line-item level	P1

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Invoice Number Format

INV-TRL-{DISTRICT_CODE}-{MSME_REF}-{SEQUENCE}

```
INV-TRL-ACC-0042-000087
    │   │   └── Zero-padded 6-digit per-MSME sequence
    │   └──────── Zero-padded 4-digit MSME programme ref
    └──────────── 3-letter uppercase district code
```

3.6 Module 5 Tax Engine [P1]

See full Ghana tax specification in Section 7. The backend must implement the tax engine as a standalone service class used by both the Sales module and the tax calculation preview endpoint.

Requirements

ID	Requirement	Priority
TAX-01	Tax engine implements three regimes: not_registered, flat_rate, standard	P1
TAX-02	All tax arithmetic uses Decimal type never float	P1
TAX-03	Tax rates stored in a database table (not hardcoded) updateable without redeployment	P1
TAX-04	GET /api/v1/tax/rates returns current rates for Flutter app to cache for offline calculation	P1
TAX-05	POST /api/v1/tax/calculate previews tax for a cart without creating a sale	P1
TAX-06	Tax calculation is deterministic same	P1

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ID	Requirement	Priority
	inputs always produce same output	
TAX-07	Tax rates update endpoint with admin-only access	P2
TAX-08	Tax rate change history audit log	P2

3.7 Module 6 Invoices & Receipts [P1]

Requirements

ID	Requirement	Priority
INV-R-01	Invoice response includes MSME TIN, business name, address, and VAT registration type	P1
INV-R-02	Receipt endpoint (GET /sales/{id}/receipt) returns printer-ready lines array	P1
INV-R-03	Receipt lines support: text_center, key_value, key_value_bold, item_line, divider, line_feed, cut	P1
INV-R-04	Receipt format configurable for 58mm (V3 PLUS) and 80mm (V2 PRO) printers	P1
INV-R-05	Receipt footer includes programme reference and GRA compliance note	P1

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ID	Requirement	Priority
INV-R-06	Electronic invoice PDF generation (email delivery)	P2
INV-R-07	GRA digital signature on invoices (requires GRA IT cooperation)	P3

3.8 Module 7 Susu (Savings) [P1]

Requirements

ID	Requirement	Priority
SUSU-01	Create a Susu scheme with name, type (individual/group), contribution amount, and frequency	P1
SUSU-02	Add members to a scheme with payout position	P1
SUSU-03	Record a contribution for a member against a scheme	P1
SUSU-04	Scheme summary shows member list, total contributions, current cycle progress	P1
SUSU-05	Member contribution history shows all payments, missed payments, and arrears	P1
SUSU-06	Frequency options: daily, weekly, biweekly, monthly	P1

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ID	Requirement	Priority
SUSU-07	Payout rotation tracking record which member received the payout each cycle	P2
SUSU-08	Collector commission calculation	P2
SUSU-09	SMS notification to members for upcoming contribution dates	P2

3.9 Module 8 Loans [P1]

Note: Phase 2 will integrate with Banks and Fintech providers (MTN MoMo, Fido, etc.) for automated loan disbursement and repayment.

Requirements

ID	Requirement	Priority
LOAN-01	Create an individual loan with principal, interest rate, interest type, and tenure	P1
LOAN-02	System generates full repayment schedule on loan creation	P1
LOAN-03	Two interest types supported: flat (on original principal) and reducing (on outstanding balance)	P1
LOAN-04	Record a loan repayment system updates outstanding balance and schedule	P1
LOAN-05	System rejects overpayments returns HTTP 422 if amount exceeds outstanding balance	P1

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ID	Requirement	Priority
LOAN-06	Loan list shows portfolio summary: total active loans, total outstanding, loans in arrears	P1
LOAN-07	All loan arithmetic uses Decimal never float	P1
LOAN-08	Group / solidarity loan support	P2
LOAN-09	Penalty calculation for late payments	P2
LOAN-10	Loan approval workflow (application → review → approve/reject)	P2
LOAN-11	Integration with Banks and Fintech providers (MTN MoMo, Fido) for loan disbursement	P2

Amortisation Flat Rate Example

Principal: GHS 500.00

Rate: 5% flat per month

Tenure: 5 months

Total interest = 500.00 × 0.05 × 5 = 125.00

Total repayable = 625.00

Monthly instalment = 625.00 / 5 = 125.00

3.10 Module 9 Expenses [P1]

New module. Expense recording is critical for producing a meaningful Business Summary (P&L). Without expenses, net profit cannot be calculated. This module is a first-class backend module, not an afterthought.

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Requirements

ID	Requirement	Priority
EXP-01	MSME or programme staff can record a business expense with category, amount, date, and description	P1
EXP-02	Expense categories are predefined but extensible: stall_rent, transport, staff_wages, utilities, supplies, loan_repayment, equipment, market_dues, packaging, other	P1
EXP-03	Expense stores payment method (cash, mobile money, bank)	P1
EXP-04	Optional receipt reference field for paper receipt tracking	P1
EXP-05	Expenses list endpoint supports filtering by date range and category	P1
EXP-06	Expense totals are used in the Business Summary report (RPT-02)	P1
EXP-07	Expenses included in the offline sync batch (expense transaction type)	P1
EXP-08	Expenses included in /sync/master-data if recorded offline on the device	P1

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ID	Requirement	Priority
EXP-09	Expenses breakdown by category for a given period (Expense Report)	P1
EXP-10	Photo capture of expense receipt (attach image to expense record)	P2
EXP-11	Recurring expense templates (e.g. monthly stall rent auto-reminds)	P2

API Endpoints

Method	Endpoint	Description
POST	/api/v1/msme/{id}/expenses	Record a new expense
GET	/api/v1/msme/{id}/expenses	List expenses (with filters)
GET	/api/v1/msme/{id}/expenses/{expense_id}	Get single expense
PATCH	/api/v1/msme/{id}/expenses/{expense_id}	Update expense
DELETE	/api/v1/msme/{id}/expenses/{expense_id}	Soft delete (admin only)

Expense Data Fields

Field	Type	Required	Notes
id	ULID	Auto	exp_{ulid}
msme_id	string	Yes	
category	enum	Yes	See category list above
amount	NUMERIC(12,2)	Yes	Never float

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Field	Type	Required	Notes
description	text	No	
expense_date	date	Yes	
payment_method	enum	No	cash, mtn_momo, vodafone_cash, bank_transfer, other
receipt_ref	string	No	Paper receipt number
recorded_by	string	Auto	User ID
client_ref	string	No	UUID for offline sync idempotency
created_at	datetime	Auto	

3.11 Module 10 Reports [P1]

Phase 1 note: Financial reports are a simplified revenue-minus-expenses Business Summary, not a full double-entry accounting system. Full double-entry P&L and Balance Sheet are Phase 2. Reports are based on the reference POS system report types, filtered to those relevant for an MSME market trader context.

Selected Reports Phase 1

The following report types are included in Phase 1, selected from the reference POS system as relevant to the programme's MSME use case:

#	Report Name	Source Module	Priority
1	Business Summary (simplified P&L)	Sales + Expenses	P1
2	Tax Report	Sales / Tax Engine	P1
3	Sales Dashboard (KPIs + chart)	Sales	P1
4	Product Sell Report	Sales + Products	P1

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#	Report Name	Source Module	Priority
5	Stock Report	Products / Inventory	P1
6	Stock Adjustment Report	Stock Movements	P1
7	Sell Payment Report	Payments	P1
8	Customer Report	Customers + Sales	P1
9	Expense Report	Expenses	P1
10	Activity Log	Audit Trail	P1
11	Purchase & Sale Summary	Sales + Stock Movements	P1
12	Profit / Loss Report (full double-entry)	Accounting Engine	P2
13	Supplier Report	Supplier module	P2
14	Product Purchase Report	Stock Movements + Suppliers	P2
15	District / Zone Overview	All MSMEs	P1

Not included (not relevant for this use case): Register Report, Sales Representative Report, Customer Groups Report, Brands Report, Warranties Report. These are standard POS features for retail chains, not applicable to individual informal market traders.

Requirements

ID	Requirement	Priority
RPT-01	**Business Summary (Simplified P&L):** Total sales, cost of goods sold, gross profit, total expenses by category, net profit for any date range	P1
RPT-02	**Tax Report:** Total tax collected by period, broken down by regime (flat rate / standard), and by tax type (VAT, NHIL,	P1

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ID	Requirement	Priority
	GETFund, COVID levy). Used to prepare GRA returns	
RPT-03	**Sales Dashboard:** KPI cards for today/week/month total revenue, transaction count, average transaction value, gross profit margin	P1
RPT-04	**Product Sell Report:** Units sold, revenue, and gross margin per product for a period. Identifies best and worst performing products	P1
RPT-05	**Stock Report:** Current stock levels, stock value, and low-stock/out-of-stock alerts across all products	P1
RPT-06	**Stock Adjustment Report:** All stock movements (restocks, wastage, adjustments) for a period with quantities, unit costs, and reasons	P1
RPT-07	**Sell Payment Report:** Sales broken down by payment method (cash, mobile money, credit, card) for a period. Shows outstanding credit total	P1
RPT-08	**Customer Report:** Customer purchase history, total spend, outstanding credit balance, and last purchase date per customer	P1
RPT-09	**Expense Report:** Expenses broken down by category and period. Month-on-month comparison	P1
RPT-10	**Activity Log:** Timestamped log of all significant system actions sales, voids, stock adjustments, user logins, profile changes for audit purposes	P1

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ID	Requirement	Priority
RPT-11	**Purchase & Sale Summary:** Side-by-side view of stock purchased (restocks) vs stock sold for a period. Shows gross movement per product	P1
RPT-12	**District / Zone Overview:** For programme staff enrolled MSMEs, active MSMEs, total transactions, total revenue, total tax collected by zone	P1
RPT-13	Daily sales chart data (last 7 or 30 days) for dashboard chart widget	P1
RPT-14	Top trending products by revenue and units sold	P1
RPT-15	Full double-entry P&L and Balance Sheet	P2
RPT-16	Cash Flow Statement	P2
RPT-17	GRA compliance dashboard (tax collected vs filed vs outstanding)	P2
RPT-18	National rollup analytics for GRA (tax gap, sector analysis)	P3

Report API Endpoints

Method	Endpoint	Report	Priority
GET	/api/v1/msme/{id}/reports/summary	Business Summary (P&L)	P1
GET	/api/v1/msme/{id}/reports/tax	Tax Report	P1
GET	/api/v1/msme/{id}/reports/dashboard	Sales Dashboard KPIs	P1
GET	/api/v1/msme/{id}/reports/products/sell	Product Sell Report	P1
GET	/api/v1/msme/{id}/reports/stock	Stock Report	P1

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Method	Endpoint	Report	Priority
GET	/api/v1/msme/{id}/reports/stock/adjustments	Stock Adjustment Report	P1
GET	/api/v1/msme/{id}/reports/payments	Sell Payment Report	P1
GET	/api/v1/msme/{id}/reports/customers	Customer Report	P1
GET	/api/v1/msme/{id}/reports/expenses	Expense Report	P1
GET	/api/v1/msme/{id}/reports/activity	Activity Log	P1
GET	/api/v1/msme/{id}/reports/purchase-sale	Purchase & Sale Summary	P1
GET	/api/v1/reports/district/{district}	District / Zone Overview	P1

All report endpoints accept ?from=YYYY-MM-DD&to=YYYY-MM-DD query parameters.

3.12 Module 11 Offline Sync [P1]

Requirements

ID	Requirement	Priority
SYNC-01	POST /sync/batch accepts an array of offline transactions and processes each atomically	P1
SYNC-02	Each transaction in the batch uses its client_ref for idempotency	P1
SYNC-03	Batch response returns per-transaction results (success/failure) partial batch success is valid	P1
SYNC-04	GET /sync/master-data/{msme_id}?since	P1

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ID	Requirement	Priority
	= returns only records changed since the timestamp	
SYNC-05	Master data includes: products, customers, tax rates, MSME profile	P1
SYNC-06	Sync batch supports transaction types: sale, stock_movement, expense, susu_contribution, loan_repayment	P1
SYNC-07	Full bidirectional conflict resolution (server wins with conflict flag)	P2

4. Component 2 Flutter POS App (Sunmi)

4.1 Overview

A Flutter Android app installed on Sunmi V3 PLUS and V2 PRO devices. Designed for non-literate market traders. Must work fully offline for the core sale-recording function.

APK distribution: Direct APK sideload not via Google Play Store for the pilot. The APK is built and distributed by the Trident Aliska team.

4.2 Screen Requirements [P1]

Screen 1 Login

ID	Requirement
UI-LOGIN-01	Username and password fields with large tap targets (minimum 56dp)
UI-LOGIN-02	Triliska logo and "Business Management Tools" title
UI-LOGIN-03	Error message displayed below form on failed login

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ID	Requirement
UI-LOGIN-04	Loading indicator during authentication
UI-LOGIN-05	"Remember me" option to avoid re-login on app restart

Screen 2 POS Sale Screen (Primary Screen)

ID	Requirement
UI-POS-01	Product grid of large tiles minimum 2 columns, tap to add to cart
UI-POS-02	Each product tile shows: product name, price in GHS, stock level indicator
UI-POS-03	Cart panel shows: items added, quantities (+/- controls), line totals
UI-POS-04	Tax breakdown shown in real time as items are added shows VAT, total
UI-POS-05	Total amount displayed prominently in large text
UI-POS-06	"Charge" button large, green, full width at bottom
UI-POS-07	Offline indicator in header shows pending sync count when offline
UI-POS-08	Customer selection (optional) search by name or phone
UI-POS-09	Products loaded from local Drift cache screen works with zero internet

Screen 3 Checkout / Payment

ID	Requirement
UI-CHK-01	Payment method selector: Cash, Mobile Money, Card, Credit

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ID	Requirement
UI-CHK-02	For cash: amount tendered input, change due display
UI-CHK-03	Confirm button triggers sale recording flow
UI-CHK-04	On success: navigate to receipt screen immediately
UI-CHK-05	On offline: save to queue, show "Sale saved syncing later", navigate to receipt

Screen 4 Receipt Screen

ID	Requirement
UI-RCT-01	Show sale summary: items, tax breakdown, total, payment method
UI-RCT-02	"Print Receipt" button triggers Sunmi thermal printer
UI-RCT-03	If API unavailable, build receipt from local Drift sale record
UI-RCT-04	"New Sale" button clears cart and returns to POS screen
UI-RCT-05	Invoice number displayed prominently

Screen 5 Inventory

ID	Requirement
UI-INV-01	List of all products with current stock, colour-coded: green (ok), amber (low), red (out)
UI-INV-02	Tap product to view details and edit price or threshold
UI-INV-03	"Add Stock" button on each product quick restock entry

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ID	Requirement
UI-INV-04	Add new product button
UI-INV-05	Stock Adjustment Report accessible from inventory screen

Screen 6 Expenses

ID	Requirement
UI-EXP-01	List of expenses for current month with category, amount, date
UI-EXP-02	"Add Expense" button large, accessible from main navigation
UI-EXP-03	Add expense form: category picker, amount, date, optional description
UI-EXP-04	Expense total for month shown as summary card at top
UI-EXP-05	Expenses recorded offline with client_ref and synced when connected

Screen 7 Susu

ID	Requirement
UI-SUSU-01	List of active Susu schemes
UI-SUSU-02	Scheme detail: member list, contribution status (paid/missed), totals
UI-SUSU-03	"Record Contribution" select member, confirm amount, payment method
UI-SUSU-04	Member history showing all contributions and missed cycles

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Screen 8 Loans

ID	Requirement
UI-LOAN-01	List of active loans with outstanding balance and next due date
UI-LOAN-02	Loan detail: full repayment schedule with status per instalment
UI-LOAN-03	"Record Repayment" enter amount and confirm
UI-LOAN-04	Create new loan form

Screen 9 Reports

ID	Requirement
UI-RPT-01	KPI cards: Today's revenue, this month's revenue, net profit
UI-RPT-02	Sales chart: bar chart of last 7 days
UI-RPT-03	Top trending products list
UI-RPT-04	Period selector: today, this week, this month, last month
UI-RPT-05	Credit outstanding total with link to customer list
UI-RPT-06	Tax summary card: tax collected this period
UI-RPT-07	Expense summary card: total expenses this period
UI-RPT-08	Stock alerts: count of low-stock and out-of-stock items

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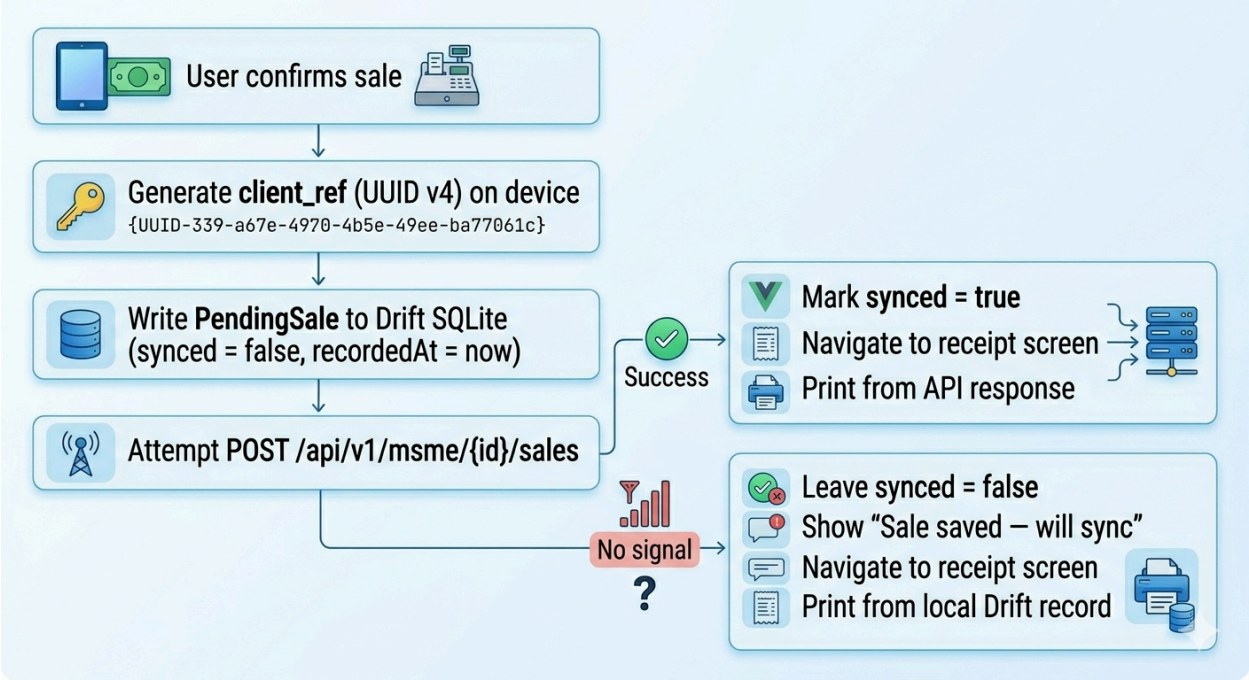
4.3 Offline Architecture [P1]

Sale Recording Flow



ID	Requirement	Priority
OFF-01	App must function fully offline for core sale recording and receipt printing	P1
OFF-02	Products, customers, tax rates cached in Drift SQLite refreshed on every sync	P1
OFF-03	All pending sales stored in Drift with synced = false flag	P1
OFF-04	Sync engine triggers automatically when network connectivity restored	P1
OFF-05	Sync engine batches pending sales and calls POST /sync/batch	P1
OFF-06	Sync status indicator always visible in app header	P1

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ID	Requirement	Priority
OFF-07	Offline storage must survive app restarts and device reboots	P1
OFF-08	Tax calculation runs locally from cached rates never blocks on API	P1
OFF-09	App handles partial sync failure gracefully failed items remain queued	P1
OFF-10	Expenses recorded offline also queued and synced via POST /sync/batch	P1
OFF-11	Full conflict resolution (multi-device editing of same record)	P2

4.4 Sunmi Hardware Integration [P1]

ID	Requirement	Priority
HW-01	Receipt printing via Sunmi Printer SDK (sunmi_printer_plus package)	P1
HW-02	App supports both 58mm (V3 PLUS) and 80mm (V2 PRO) printer widths	P1
HW-03	Receipt maps API lines array to printer SDK calls (text, key-value, divider, cut)	P1
HW-04	Printer errors (paper out, not connected) shown as user-friendly alert	P1
HW-05	Barcode scanner input (V2 PRO) populates product search on POS screen	P1
HW-06	Customer pole display (V2 PRO) shows cart total during checkout	P2
HW-07	NFC payment tap (V2 PRO)	P2

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4.5 UI Design Requirements [P1]

ID	Requirement	Priority
DES-01	All interactive elements minimum 56dp height	P1
DES-02	Body text minimum 16sp	P1
DES-03	High contrast WCAG AA minimum	P1
DES-04	Monetary amounts always displayed as "GHS X.XX"	P1
DES-05	Status indicators use colour + icon never colour alone	P1
DES-06	Loading states shown for all async operations	P1
DES-07	Error states handled gracefully never show stack traces	P1
DES-08	English language for Phase 1 pilot	P1
DES-09	Twi language support	P2
DES-10	Ga, Ewe, Hausa, Dagbani language support	P2

Brand colours:

Token	Hex	Usage
navy	#1A3A5C	Headers, primary buttons
blue	#1E6BA8	Icons, links, secondary actions
lightBlue	#D6E8F7	Card backgrounds, tints

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Token	Hex	Usage
green	#1A7A4A	Success, paid, positive status
amber	#F59E0B	Warning, offline, pending
red	#B91C1C	Error, missed, overdue
background	#F2F5F8	App background
surface	#FFFFFF	Cards, inputs

---

5. Component 3 GRA Middleware

5.1 Overview

A standalone FastAPI microservice owned **exclusively by Aliska (Trident Aliska Support)**. This service is the sole gateway to GRA systems and payment gateways. It is called only by triliska-backend via an internal IP-whitelisted API.

The Vietnam/India development team has zero access to this repository.

5.2 Security Architecture [P1]

ID	Requirement	Priority
MW-SEC-01	Service is NOT publicly accessible internal network only	P1
MW-SEC-02	All requests must include X-Internal-Key header matching environment variable	P1
MW-SEC-03	IP whitelist only accept requests from triliska-backend server IPs	P1

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ID	Requirement	Priority
MW-SEC-04	All GRA credentials stored in environment variables never in code or repository	P1
MW-SEC-05	Audit log every GRA API call: system, endpoint, payload hash, response status, duration	P1
MW-SEC-06	Log payload HASH only never the raw payload (contains sensitive tax data)	P1
MW-SEC-07	All GRA connections use HTTPS with SSL certificate verification never disabled	P1

5.3 Module 1 Tax Return Generation [P1]

Requirements

ID	Requirement	Priority
MW-TAX-01	Accept MSME ID and period, aggregate all sales from backend DB	P1
MW-TAX-02	Calculate: total_sales, taxable_sales, vat_collected, nhil_collected, getfund_collected, covid_levy_collected	P1
MW-TAX-03	Generate GRA VAT 3 format PDF using WeasyPrint	P1
MW-TAX-04	PDF includes all mandatory GRA fields (see Section 7.3)	P1

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ID	Requirement	Priority
MW-TAX-05	Store generated PDF in object storage, return URL	P1
MW-TAX-06	Write audit record for every PDF generated	P1
MW-TAX-07	Electronic submission to GRA MTS API	P2
MW-TAX-08	Store GRA filing reference number and confirmation	P2
MW-TAX-09	Retry logic with exponential backoff for failed submissions	P2

VAT 3 Form Mandatory Sections

Section A Taxpayer Information: TIN, business name, address, tax period, VAT registration type
Section B Output Tax: Total taxable sales, VAT rate, VAT on sales, NHIL (standard rate), GETFund levy (standard rate), COVID-19 levy (standard rate)
Section C Input Tax: Total purchases, input VAT claimable *(standard rate only not applicable for pilot flat-rate MSMEs)*
Section D Net Tax Payable: Output tax minus input tax credit
Section E Declaration: Declaration statement, signature line, name, date, designation

5.4 Module 2 GRA Compliance Check [P2]

GET /internal/compliance/{tin}
→ Query GRA GITMS for MSME compliance status
→ Return: registration_status, compliance_status, last_return_filed,

outstanding_returns, outstanding_balance

5.5 Module 3 Payments [P2]

POST /internal/payments/initiate
→ Initiate mobile money payment for tax obligations
→ Supports: MTN MoMo, Vodafone Cash, AirtelTigo
→ Returns: payment_id, gateway_ref, status

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5.6 Module 4 Compliance Alert Scheduler [P1 stub / P2 live]

ID	Requirement	Priority
MW-ALERT-01	Celery beat scheduler runs daily at 08:00 Ghana time (UTC+0)	P1
MW-ALERT-02	Check all MSMEs for VAT filing deadlines	P1
MW-ALERT-03	Log reminder intent to file (stub for pilot no SMS sent)	P1
MW-ALERT-04	Send SMS 7 days and 1 day before filing deadline	P2
MW-ALERT-05	SMS gateway configurable via environment variable	P2

5.7 Internal API Contract

Endpoint	Phase	Description
POST /internal/tax-return/generate	P1	Generate VAT 3 PDF
POST /internal/tax-return/submit	P2	Submit to GRA MTS API
GET /internal/compliance/{tin}	P2	Check MSME compliance via GITMS
POST /internal/payments/initiate	P2	Initiate mobile money payment

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Endpoint	Phase	Description
GET /internal/health	P1	Health check

6. Database Schema

6.1 Core Tables triliska-backend

```
-- Users and Authentication

CREATE TABLE users (

   id            VARCHAR(30) PRIMARY KEY,
   username       VARCHAR(100) UNIQUE NOT NULL,
   email          VARCHAR(255) UNIQUE,
   hashed_password VARCHAR(255) NOT NULL,
   role           VARCHAR(30) NOT NULL,
   msme_id        VARCHAR(30),
   district       VARCHAR(100),
   zone           VARCHAR(100),
   is_active      BOOLEAN DEFAULT TRUE,

   created_at     TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

CREATE TABLE refresh_tokens (
   id         VARCHAR(30) PRIMARY KEY,
   user_id    VARCHAR(30) REFERENCES users(id),
   token_hash  VARCHAR(255) UNIQUE NOT NULL,
   expires_at  TIMESTAMP WITH TIME ZONE NOT NULL,
   revoked    BOOLEAN DEFAULT FALSE,
   created_at  TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- MSME Profiles

CREATE TABLE msme_profiles (

   id              VARCHAR(30) PRIMARY KEY,
```

```
    business_name       VARCHAR(255) NOT NULL,
    owner_name          VARCHAR(255) NOT NULL,
    owner_phone         VARCHAR(20) NOT NULL,
    owner_national_id   VARCHAR(50) NOT NULL,
    tin                 VARCHAR(50),
    business_type       VARCHAR(30) NOT NULL,
    sector              VARCHAR(50) NOT NULL,
    address             TEXT NOT NULL,
    gps_lat             NUMERIC(10, 7),
    gps_lng             NUMERIC(10, 7),
    district            VARCHAR(100) NOT NULL,
    zone                VARCHAR(100),
    vat_status          VARCHAR(30) NOT NULL DEFAULT 'not_registered',
    annual_turnover_band VARCHAR(30) NOT NULL DEFAULT 'below_200k',
    threshold_warning   BOOLEAN DEFAULT FALSE,
    preferred_language  VARCHAR(5) DEFAULT 'en',
    programme_ref       VARCHAR(20) UNIQUE NOT NULL,
    registration_date   TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    status              VARCHAR(20) DEFAULT 'active',
    programme_staff_id  VARCHAR(30) REFERENCES users(id),
    created_at          TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    updated_at          TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

-- Products and Inventory

CREATE TABLE products (

    id                  VARCHAR(30) PRIMARY KEY,
    msme_id             VARCHAR(30) NOT NULL REFERENCES msme_profiles(id),
    name                VARCHAR(255) NOT NULL,
    sku                 VARCHAR(100),
    barcode             VARCHAR(100),
    category            VARCHAR(100),
    unit                VARCHAR(50),
    selling_price       NUMERIC(12, 2) NOT NULL,
    cost_price          NUMERIC(12, 2),
    tax_applicable      BOOLEAN DEFAULT TRUE,
    current_stock       INTEGER DEFAULT 0,
    low_stock_threshold INTEGER DEFAULT 5,
    is_active           BOOLEAN DEFAULT TRUE,
```

```
    created_at        TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    updated_at        TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

CREATE TABLE stock_movements (

    id          VARCHAR(30) PRIMARY KEY,
    product_id    VARCHAR(30) NOT NULL REFERENCES products(id),
    msme_id       VARCHAR(30) NOT NULL,
    movement_type VARCHAR(30) NOT NULL,
    quantity      INTEGER NOT NULL,
    unit_cost     NUMERIC(12, 2),
    notes         TEXT,

    reference     VARCHAR(100),
    recorded_by   VARCHAR(30) REFERENCES users(id),
    created_at    TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- Sales

CREATE TABLE sales (
    id              VARCHAR(30) PRIMARY KEY,
    client_ref       VARCHAR(36) UNIQUE NOT NULL,
    msme_id          VARCHAR(30) NOT NULL REFERENCES msme_profiles(id),
    customer_id       VARCHAR(30),
    sale_type        VARCHAR(30) NOT NULL,
    status           VARCHAR(20) DEFAULT 'completed',
    invoice_number    VARCHAR(50) UNIQUE NOT NULL,
    subtotal         NUMERIC(12, 2) NOT NULL,
    discount_total    NUMERIC(12, 2) DEFAULT 0,
    taxable_amount    NUMERIC(12, 2) NOT NULL,
    vat_amount        NUMERIC(12, 2) DEFAULT 0,
    nhil_amount       NUMERIC(12, 2) DEFAULT 0,
    getfund_amount    NUMERIC(12, 2) DEFAULT 0,
    covid_levy_amount NUMERIC(12, 2) DEFAULT 0,
    total_tax        NUMERIC(12, 2) DEFAULT 0,
    total_amount      NUMERIC(12, 2) NOT NULL,
    tax_regime       VARCHAR(30) NOT NULL,
    notes            TEXT,
    voided_at        TIMESTAMP WITH TIME ZONE,
```

```
    voided_by        VARCHAR(30),
    void_reason       TEXT,
    device_id        VARCHAR(100),
    recorded_at       TIMESTAMP WITH TIME ZONE NOT NULL,
    created_at       TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

CREATE TABLE sale_items (

    id            VARCHAR(30) PRIMARY KEY,
    sale_id        VARCHAR(30) NOT NULL REFERENCES sales(id),
    product_id      VARCHAR(30) NOT NULL,
    product_name    VARCHAR(255) NOT NULL,
    quantity        INTEGER NOT NULL,
    unit_price      NUMERIC(12, 2) NOT NULL,
    discount_amount NUMERIC(12, 2) DEFAULT 0,
    line_subtotal   NUMERIC(12, 2) NOT NULL,
    tax_amount      NUMERIC(12, 2) DEFAULT 0,
    line_total      NUMERIC(12, 2) NOT NULL

);

CREATE TABLE payments (

    id            VARCHAR(30) PRIMARY KEY,
    sale_id        VARCHAR(30) NOT NULL REFERENCES sales(id),
    method          VARCHAR(30) NOT NULL,
    amount          NUMERIC(12, 2) NOT NULL,
    amount_tendered  NUMERIC(12, 2),
    change_due       NUMERIC(12, 2),
    mobile_money_ref VARCHAR(100),
    paid_at         TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

-- Customers

CREATE TABLE customers (

    id        VARCHAR(30) PRIMARY KEY,
    msme_id    VARCHAR(30) NOT NULL REFERENCES msme_profiles(id),
```

```
    name        VARCHAR(255) NOT NULL,
    phone       VARCHAR(20),
    address     TEXT,
    notes       TEXT,
    created_at  TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

-- Tax Configuration

CREATE TABLE tax_config (

    id              SERIAL PRIMARY KEY,
    vat_standard    NUMERIC(6, 4) NOT NULL DEFAULT 0.1500,
    vat_flat_rate   NUMERIC(6, 4) NOT NULL DEFAULT 0.0300,
    nhil            NUMERIC(6, 4) NOT NULL DEFAULT 0.0250,
    getfund         NUMERIC(6, 4) NOT NULL DEFAULT 0.0250,
    covid_levy      NUMERIC(6, 4) NOT NULL DEFAULT 0.0100,
    threshold_lower NUMERIC(12, 2) NOT NULL DEFAULT 500000.00,
    threshold_upper NUMERIC(12, 2) NOT NULL DEFAULT 750000.00,
    effective_date  DATE NOT NULL,
    created_by      VARCHAR(30),
    created_at      TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

-- Susu

CREATE TABLE susu_schemes (

    id                VARCHAR(30) PRIMARY KEY,
    msme_id           VARCHAR(30) NOT NULL,
    scheme_name       VARCHAR(255) NOT NULL,
    scheme_type       VARCHAR(20) NOT NULL,
    contribution_amount NUMERIC(12, 2) NOT NULL,
    frequency         VARCHAR(20) NOT NULL,
    start_date        DATE NOT NULL,
    collector_name    VARCHAR(255),
    collector_phone   VARCHAR(20),
    notes             TEXT,
    status            VARCHAR(20) DEFAULT 'active',
    created_at        TIMESTAMP WITH TIME ZONE DEFAULT NOW()
```

```
);

CREATE TABLE susu_members (

    id            VARCHAR(30) PRIMARY KEY,
    scheme_id      VARCHAR(30) NOT NULL REFERENCES susu_schemes(id),
    msme_id_ref     VARCHAR(30),
    member_name     VARCHAR(255) NOT NULL,
    member_phone    VARCHAR(20),
    payout_position INTEGER,
    joined_date     DATE NOT NULL,
    status         VARCHAR(20) DEFAULT 'active'

);

CREATE TABLE susu_contributions (

    id              VARCHAR(30) PRIMARY KEY,
    scheme_id        VARCHAR(30) NOT NULL REFERENCES susu_schemes(id),
    member_id        VARCHAR(30) NOT NULL REFERENCES susu_members(id),
    amount           NUMERIC(12, 2) NOT NULL,
    contribution_date DATE NOT NULL,
    payment_method    VARCHAR(30),
    client_ref       VARCHAR(36),
    notes            TEXT,
    recorded_at      TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

-- Loans

CREATE TABLE loans (

    id                VARCHAR(30) PRIMARY KEY,
    msme_id             VARCHAR(30) NOT NULL,
    borrower_name        VARCHAR(255) NOT NULL,
    borrower_phone       VARCHAR(20),
    msme_id_ref          VARCHAR(30),
    principal           NUMERIC(12, 2) NOT NULL,
    interest_rate        NUMERIC(6, 2) NOT NULL,
    interest_type        VARCHAR(20) NOT NULL,
```

```
    tenure_months        INTEGER NOT NULL,
    total_interest       NUMERIC(12, 2) NOT NULL,
    total_repayable      NUMERIC(12, 2) NOT NULL,
    monthly_instalment   NUMERIC(12, 2) NOT NULL,
    balance_outstanding  NUMERIC(12, 2) NOT NULL,
    disbursement_date    DATE NOT NULL,
    repayment_start_date DATE NOT NULL,
    purpose              TEXT,
    notes                TEXT,
    status               VARCHAR(20) DEFAULT 'active',
    created_at           TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

CREATE TABLE loan_schedule (

    id                   VARCHAR(30) PRIMARY KEY,
    loan_id              VARCHAR(30) NOT NULL REFERENCES loans(id),
    instalment_number    INTEGER NOT NULL,
    due_date             DATE NOT NULL,
    principal_component  NUMERIC(12, 2) NOT NULL,
    interest_component   NUMERIC(12, 2) NOT NULL,
    instalment_amount    NUMERIC(12, 2) NOT NULL,
    balance_after        NUMERIC(12, 2) NOT NULL,
    status               VARCHAR(20) DEFAULT 'upcoming'

);

CREATE TABLE loan_repayments (

    id                   VARCHAR(30) PRIMARY KEY,
    loan_id              VARCHAR(30) NOT NULL REFERENCES loans(id),
    amount               NUMERIC(12, 2) NOT NULL,
    payment_method       VARCHAR(30),
    payment_date         DATE NOT NULL,
    principal_applied    NUMERIC(12, 2) NOT NULL,
    interest_applied     NUMERIC(12, 2) NOT NULL,
    balance_before       NUMERIC(12, 2) NOT NULL,
    balance_after        NUMERIC(12, 2) NOT NULL,
    notes                TEXT,
    mobile_money_ref     VARCHAR(100),
    client_ref           VARCHAR(36),
```

```
    recorded_at       TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

-- Expenses (new module v1.1)

CREATE TABLE expenses (

    id            VARCHAR(30) PRIMARY KEY,
    msme_id       VARCHAR(30) NOT NULL REFERENCES msme_profiles(id),
    category      VARCHAR(50) NOT NULL,

    amount        NUMERIC(12, 2) NOT NULL,
    description   TEXT,
    expense_date  DATE NOT NULL,
    payment_method VARCHAR(30),
    receipt_ref   VARCHAR(100),
    client_ref    VARCHAR(36),
    recorded_by   VARCHAR(30) REFERENCES users(id),
    created_at    TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);

-- Audit Log

CREATE TABLE audit_log (
    id          BIGSERIAL PRIMARY KEY,
    entity_type VARCHAR(50),
    entity_id   VARCHAR(30),
    action      VARCHAR(50),
    changed_by  VARCHAR(30),
    old_values  JSONB,
    new_values  JSONB,
    ip_address  VARCHAR(50),
    device_id   VARCHAR(100),
    created_at  TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);
```

6.2 Middleware Tables triliska-middleware

```
CREATE TABLE tax_returns (
```

```
    id                  VARCHAR(30) PRIMARY KEY,
    msme_id             VARCHAR(30) NOT NULL,
    tin                 VARCHAR(50),
    period_start        DATE NOT NULL,
    period_end          DATE NOT NULL,
    return_type         VARCHAR(20) DEFAULT 'vat',
    total_sales         NUMERIC(12, 2),
    taxable_sales       NUMERIC(12, 2),
    vat_collected       NUMERIC(12, 2),
    nhil_collected      NUMERIC(12, 2),
    getfund_collected   NUMERIC(12, 2),
    covid_levy_collected NUMERIC(12, 2),
    total_tax_payable   NUMERIC(12, 2),
    pdf_path            TEXT,
    status              VARCHAR(30) DEFAULT 'generated',
    gra_submission_status VARCHAR(30) DEFAULT 'pending_manual',
    gra_reference       VARCHAR(100),

    generated_at        TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    submitted_at        TIMESTAMP WITH TIME ZONE

);

CREATE TABLE gra_audit_log (

    id              BIGSERIAL PRIMARY KEY,
    system          VARCHAR(30) NOT NULL,
    endpoint        VARCHAR(255) NOT NULL,
    method          VARCHAR(10) NOT NULL,
    msme_id         VARCHAR(30),
    payload_hash    VARCHAR(64) NOT NULL,
    response_status INTEGER,
    response_summary TEXT,
    duration_ms     INTEGER,
    success         BOOLEAN NOT NULL,
    logged_at       TIMESTAMP WITH TIME ZONE DEFAULT NOW()

);
```

7. Ghana Tax Requirements

7.1 Tax Regimes

Regime	Turnover	VAT	NHIL	GETFund	COVID Levy
not_registered	< GHS 200,000	0%	0%	0%	0%
flat_rate	GHS 200k – 500k	3% (gross)	0%	0%	0%
standard	> GHS 500,000	15%	2.5%	2.5%	1%

Threshold flags: System flags at GHS 500,000 (approaching flat rate ceiling) and GHS 750,000 (approaching standard rate threshold). Does NOT auto-change status.

7.2 Calculation Rules

For standard rate: NHIL, GETFund, and COVID Levy are all calculated on the **VAT-exclusive amount** not on the VAT amount itself.

Standard rate, GHS 1,000.00 sale:

```
taxable_amount  = 1,000.00

VAT @ 15%       =   150.00   (1000 × 0.15)
NHIL @ 2.5%     =    25.00   (1000 × 0.025) ← on taxable, not VAT
GETFund @ 2.5%  =    25.00   (1000 × 0.025) ← on taxable, not VAT
COVID Levy @ 1% =    10.00   (1000 × 0.01)  ← on taxable, not VAT
──────────────────────────────────────────
Total tax       =   210.00

Invoice total   = 1,210.00
```

Flat rate, GHS 1,000.00 sale:

```
taxable_amount  = 1,000.00
VAT flat @ 3%   =    30.00   (1000 × 0.03)
──────────────────────────────────────────

Total tax       =    30.00
Invoice total   = 1,030.00
```

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7.3 Applicable Laws

Law	Relevance
Value Added Tax Act, 2013 (Act 870)	VAT rates, flat rate scheme, invoice requirements
Revenue Administration Act, 2016 (Act 915)	GRA operations, filing requirements
Income Tax Act, 2015 (Act 896)	Income tax for MSMEs and sole proprietors
Electronic Transactions Act, 2008 (Act 772)	Legal validity of electronic invoices
Data Protection Act, 2012 (Act 843)	Data storage, residency, security

8. Security Requirements

ID	Requirement	Priority
SEC-01	All API communication over HTTPS/TLS 1.2+ no HTTP in any environment	P1
SEC-02	Passwords stored as bcrypt hashes with cost factor ≥ 12	P1
SEC-03	JWT access tokens expire after 24 hours	P1
SEC-04	Refresh tokens stored as hashed values in database	P1
SEC-05	All database queries use parameterised statements no string interpolation	P1
SEC-06	MSME data isolated by role msme_owner cannot access other MSMEs' data	P1
SEC-07	All financial record changes written to audit_log	P1
SEC-08	MSME data stored on servers in Ghana or under Ghana-jurisdiction hosting	P1

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ID	Requirement	Priority
SEC-09	Database not publicly accessible backend is the only entry point	P1
SEC-10	Automated daily backups retained for 30 days minimum	P1
SEC-11	SSL certificate active and valid (Let's Encrypt acceptable for pilot)	P1
SEC-12	GRA API credentials stored only in environment variables in middleware	P1
SEC-13	Rate limiting on auth endpoints (max 10 requests/minute per IP)	P1
SEC-14	Multi-factor authentication for admin and district_officer	P2
SEC-15	Penetration test before national scale-up	P3

9. Non-Functional Requirements

9.1 Performance

ID	Requirement	Target	Priority
PERF-01	POST /sales response time	< 500ms at p95	P1
PERF-02	GET /reports/dashboard response time	< 2s at p95	P1
PERF-03	POST /sync/batch (50 transactions)	< 3s at p95	P1
PERF-04	Concurrent users for pilot	50 simultaneous	P1

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ID	Requirement	Target	Priority
PERF-05	Concurrent users for national scale	2,000,000 registered, 50,000 simultaneous	P3

9.2 Availability

ID	Requirement	Target	Priority
AVAIL-01	API uptime during pilot	99%	P1
AVAIL-02	Planned maintenance window	Off-peak hours (01:00–04:00 GMT)	P1
AVAIL-03	API uptime at national scale	99.9%	P3

9.3 Data Integrity

ID	Requirement	Priority
DATA-01	All financial transactions ACID-compliant no partial writes	P1
DATA-02	Posted sales cannot be deleted only voided with audit trail	P1
DATA-03	Tax calculations are deterministic and reproducible from stored data	P1
DATA-04	All monetary arithmetic uses Decimal zero tolerance for float	P1

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10. Acceptance Criteria Phase 1 Pilot Launch

Each item is a binary pass/fail. All must pass before launch at Makola Market.

Backend

- ☐ POST /auth/login returns JWT for valid credentials

- ☐ POST /msme creates and returns a new MSME profile with Ghana Card field
- ☐ POST /msme/{id}/products creates a product with correct stock
- ☐ POST /msme/{id}/sales creates a sale with correct Ghana tax calculation (verified against Section 7.2 worked examples)
- ☐ Duplicate client_ref returns HTTP 200 with original sale not a duplicate
- ☐ GET /msme/{id}/sales/{id}/receipt returns printer-ready lines array
- ☐ POST /msme/{id}/expenses creates an expense and it appears in business summary
- ☐ GET /msme/{id}/reports/summary includes expenses in net profit calculation
- ☐ GET /msme/{id}/reports/tax returns correct tax collected for period
- ☐ GET /msme/{id}/reports/products/sell returns sales per product
- ☐ GET /msme/{id}/reports/stock/adjustments returns stock movements
- ☐ POST /msme/{id}/susu/schemes and contribution recording work end-to-end
- ☐ POST /msme/{id}/loans generates correct repayment schedule
- ☐ POST /sync/batch processes 10 offline sales and 5 offline expenses with no duplicates
- ☐ MSME owner cannot access another MSME's data (verified by test)
- ☐ All endpoints return HTTP 401 without valid JWT
- ☐ System is live at https://api.triliska.com with valid SSL

Flutter App (Hybrid)

- ☐ App installs via APK sideload on Sunmi V3 PLUS
- ☐ Login screen works and stores JWT securely
- ☐ POS screen loads products from cache works with WiFi off
- ☐ Adding items to cart shows correct tax calculation in real time
- ☐ Completing a cash sale records the transaction (online and offline)
- ☐ Receipt prints correctly on Sunmi 58mm printer
- ☐ Offline sale queues locally and syncs when WiFi restored no duplicate
- ☐ Expenses screen allows recording an expense (online and offline)
- ☐ Offline expense queues and syncs correctly
- ☐ Sync status indicator shows correct state (online/offline/pending count)
- ☐ Susu contribution can be recorded
- ☐ Loan repayment can be recorded
- ☐ Reports screen shows correct KPIs, tax summary, and expense summary

Middleware

- ☐ POST /internal/tax-return/generate produces a GRA-compliant VAT 3 PDF
- ☐ PDF includes MSME TIN, period, tax breakdown, and declaration section

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- ☐ Audit log entry written for every call
- ☐ Service only responds to requests with valid X-Internal-Key
- ☐ Health check returns healthy status

11. Phase 2 Requirements (Planned)

Phase 2 targets 4–8 weeks post-pilot launch, subject to GRA API access being granted.

Area	Requirement
GRA Integration	Electronic VAT return submission via GRA MTS API
GRA Integration	MSME TIN verification via GITMS
GRA Integration	Revenue notification to TRIPS
Payments	MTN MoMo payment for tax obligations
Payments	Vodafone Cash and AirtelTigo support
Loans	Bank and Fintech integration (MTN MoMo, Fido) for loan disbursement
Offline	Full offline sync with conflict resolution
Offline	Offline mode survives 72 hours without connectivity
Hardware	Sunmi V2 PRO pole display support
Language	Twi language support across all screens
Reporting	Full double-entry P&L and Balance Sheet
Reporting	Cash Flow Statement
Reporting	Supplier Report and Product Purchase Report
Reporting	GRA compliance dashboard (district-level)
Loans	Group / solidarity loan support
Loans	Penalty calculation for late repayments
Susu	Payout rotation tracking

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Area	Requirement
Susu	SMS reminders for contribution dates
Security	MFA for admin and district_officer roles
Products	Product variations (size, colour, weight)
Expenses	Photo capture of expense receipts

12. Phase 3 Requirements (Planned)

Phase 3 targets national rollout 3–6 months post-pilot.

Area	Requirement
Language	Ga, Ewe, Hausa, Dagbani support
Analytics	National GRA tax gap estimation dashboard
Analytics	Sector-level compliance analysis
Analytics	Revenue forecasting model
Accounting	Full double-entry accounting engine
Accounting	Bank reconciliation
Payroll	PAYE payroll module
Scale	Infrastructure to support 2,000,000 registered MSMEs
Scale	Multi-region database for national coverage
Security	Penetration test and security audit
AI	AI-assisted tax advisory for MSMEs
Integrations	NIA (National Identification Authority)
Integrations	GRA (expanded TRIPS, corporate tax)
Integrations	RGD-ORC (Registrar General's Department)
Integrations	SSNIT (Social Security and National Insurance Trust)

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Area	Requirement
Integrations	PPA (Public Procurement Authority)
Integrations	Utilities (ECG, GWCL)
Integrations	Bank of Ghana / financial institutions

13. Glossary

Term	Definition
MSME	Micro, Small and Medium Enterprise
GRA	Ghana Revenue Authority
MTS	Modified Tax System GRA's digital tax filing platform
GITMS	Ghana Integrated Tax Management Information System
TRIPS	Total Revenue Integrated Processing System
NIA	National Identification Authority Ghana Card issuer
RGD-ORC	Registrar General's Department Online Registration Centre
SSNIT	Social Security and National Insurance Trust
TIN	Tax Identification Number issued by GRA
VAT	Value Added Tax 15% standard rate or 3% flat rate in Ghana
NHIL	National Health Insurance Levy 2.5%
GETFund	Ghana Education Trust Fund levy 2.5%
COVID Levy	COVID-19 Health Recovery Levy 1%
Susu	Rotating savings and credit scheme common in West Africa
NSP	National Service Personnel deployed as programme staff
YEA	Youth Employment Agency source of Technical Field Officers

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Term	Definition
PCT	Programme Coordinating Team Trident Aliska district management
Sunmi V3 PLUS	Android POS device with 58mm thermal printer
Sunmi V2 PRO	Android POS device with 80mm thermal printer and barcode scanner
ULID	Universally Unique Lexicographically Sortable Identifier
JWT	JSON Web Token used for API authentication
ACID	Atomicity, Consistency, Isolation, Durability database transaction properties
client_ref	Device-generated UUID on every transaction enables idempotent offline sync
flat_rate MSME	MSME registered under Ghana VAT flat rate scheme (3% on gross sales)
standard MSME	MSME registered under standard Ghana Modified Tax System (15%)

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